

House Price Prediction using Linear Regression

This project focuses on building a machine learning model to predict house prices using Linear Regression. The aim is to understand how various housing features influence price and to provide real-time price predictions.

Dataset Description

The dataset consists of housing-related attributes such as area, number of bedrooms, bathrooms, parking availability, furnishing status, and amenities like air conditioning and basement. The target variable is the house price.

Data Preprocessing

Data preprocessing involved handling missing values, encoding categorical variables using one-hot encoding, and scaling numerical features using StandardScaler to improve model performance.

Model Development

A Linear Regression model was implemented using the Scikit-learn library. The dataset was split into training and testing sets to evaluate generalization performance.

Model Evaluation

The model achieved an R-squared score of approximately 0.65, indicating that 65% of the variance in house prices is explained by the model. Mean Squared Error (MSE) was used as the primary error metric, and RMSE was considered for interpretability.

Real-Time Prediction

For real-time predictions, user inputs are transformed using the same preprocessing steps applied during training. Feature alignment ensures consistency between training and inference.

Conclusion

This project demonstrates a complete end-to-end machine learning workflow, from data preprocessing to real-time prediction. The model provides a strong baseline and can be further improved using advanced regression techniques.